TEXORA · COURSE NOTES · LECTURE SHEET

Photochemistry

Chemistry driven by light — the two laws of photochemistry, quantum yield, fluorescence and phosphorescence, and what light does to dyes and fibres.

CHE-1105 · CHEMISTRY-I · TOPIC 9 · LEVEL 1 TERM 1

1. Reactions that need a photon

Thermal reactions draw their activation energy from molecular collisions; **photochemical** reactions draw it from absorbed light — one photon at a time. Because a photon's energy is set by its wavelength ($E = h\nu = hc/\lambda$), light hands the chemist a reagent of tunable energy: ultraviolet photons rival bond energies, which is why UV drives chemistry that visible light cannot. The subject rests on two laws:

- **Grotthuss–Draper law:** only the light that is *absorbed* can produce chemical change. Transmitted or reflected light is photochemically idle — a reaction mixture must be coloured (or UV-absorbing) at the wavelength used.
- **Stark–Einstein law:** in the primary step, each absorbed photon activates exactly one molecule. (What the activated molecule then does — including starting a chain that consumes thousands more — is the secondary process, outside the law.)

The energy carried by one mole of photons (an *einstein*) is $E = N_A hc/\lambda$: at 400 nm it is about 299 kJ mol^{-1}, at 700 nm about 171 kJ mol^{-1} — the arithmetic that explains why the violet end of the spectrum bleaches, fades and degrades, and the red end mostly does not.

2. How much light is absorbed: the Lambert–Beer law

Absorbance grows with path length and concentration:

$$A = \log_{10}(I_0 / I) = \varepsilon \cdot c \cdot l$$

where ε is the molar absorption coefficient, c the concentration and l the path length. This is the measuring law of all colorimetry — the dyer's spectrophotometer reads dye concentrations, bath exhaustion (Topic 6) and shade differences through it daily. It holds well for dilute solutions and fails, instructively, when molecules associate or the light is not monochromatic.

3. Quantum yield

The efficiency of a photochemical process is its **quantum yield**:

$$\varphi = \text{(number of molecules reacted)} / \text{(number of photons absorbed)}$$

The Stark–Einstein law makes $\varphi = 1$ the naive expectation, and reality spreads widely around it. $\varphi \gg 1$ (thousands or millions) signals a **chain reaction**: one photon creates a radical that propagates through many cycles before termination — the H_2 + Cl_2 reaction is the textbook case. $\varphi \ll 1$ means most activated molecules lose their energy (by collision, fluorescence, or recombination of the fragments) before reacting, as in many photodissociations in solution where the solvent cage reunites the fragments. Measuring φ therefore reads the mechanism: the number itself testifies whether chains, cages or clean single steps are at work.

4. What excited molecules do: the Jablonski picture

Absorption lifts a molecule to an excited electronic state in about a femtosecond. From there the energy leaves by competing paths, conventionally drawn in the **Jablonski diagram**: the molecule may shed energy as heat (internal conversion, radiationless decay between states of the same spin; intersystem crossing to a triplet state), or re-emit it as light — **fluorescence**, the prompt emission (nanoseconds) from the singlet excited state back to the ground state, at longer wavelength than the absorbed light; and **phosphorescence**, the slow emission (milliseconds to seconds — the glow that lingers) from the long-lived triplet state after intersystem crossing. Or the molecule may react — dissociation, isomerisation, electron transfer — which is where photochemistry proper begins. Competing with all of these, the excited molecule may simply transfer its energy to a neighbour (**photosensitisation**: the basis of photodynamic processes and of much dye-fading chemistry, where the dye sensitises oxygen attack on the fibre).

5. Two great applications

- **Photosynthesis:** the planetary photochemical reaction — chlorophyll absorbs red and blue photons, and a chain of electron transfers builds carbohydrate from CO_2 and water; its overall quantum requirement (several photons per CO_2 fixed) reflects a multi-step, non-chain mechanism.
- **Photography (classical):** photons reduce grains of silver bromide in an emulsion to specks of silver — a latent image of a few atoms per grain — which development then amplifies enormously: an effective quantum yield engineered into the billions, and the historical model of photographic speed.

6. The textile thread: light as enemy and tool

No topic in this course meets the customer more directly. Sunlight **fades dyes** (photochemical attack on the chromophore, often sensitised) — hence the industry's **light-fastness** grading of every dyeing on

the 1–8 scale against standard blue-wool references; and it **tenders fibres**: UV breaks chains in cotton, silk and nylon, so curtains and outdoor fabrics lose strength first where the sun strikes — the photochemistry of §1 with E ≈ 300 kJ mol^{-1} photons snapping bonds of comparable energy. The defences are equally photochemical: UV-absorber finishes that sacrifice themselves by absorbing photons and shedding the energy as heat, pigments that screen, and dye selection by fastness rating. And light is also a tool: UV curing of prints and finishes, and the spectrophotometer's Lambert–Beer arithmetic, run the modern colour laboratory.

7. Glossary — key terms of this topic

TERM	MEANING
Quantum yield (φ)	Molecules reacted per photon absorbed.
Einstein	One mole of photons; its energy is $N_A hc/\lambda$.
Absorbance	$A = \log (I_0/I) = \varepsilon cl$ — the measure of light absorbed (Lambert–Beer law).
Fluorescence	Prompt light emission from an excited singlet state as it returns to the ground state.
Phosphorescence	Slow light emission from a long-lived triplet excited state.
Intersystem crossing	Radiationless transition between states of different spin (singlet → triplet).
Photosensitisation	Reaction caused in one substance by energy transferred from a light-absorbing partner.
Light fastness	A dyeing's resistance to fading by light, graded 1 (poor) to 8 (excellent).

8. Key points and summary

- Only absorbed light acts (Grotthuss–Draper), and one photon activates one molecule in the primary step (Stark–Einstein); photon energy $E = hc/\lambda$ reaches bond energies in the ultraviolet.
- Absorbance obeys Lambert–Beer ($A = \varepsilon cl$) — the working law of colour measurement — and quantum yield diagnoses mechanisms: $\varphi \gg 1$ chains, $\varphi \ll 1$ wasteful deactivation.
- Excited states decay by heat, fluorescence, phosphorescence, energy transfer or reaction; every textile light problem — fading, tendering, UV protection, light-fastness grading — is one of these paths winning.

9. Practice

Answer in your own words; brief model answers follow. ($h = 6.626 \times 10^{-34}$ J s, $c = 3.00 \times 10^{8}$ m s^{-1}, $N_A = 6.022 \times 10^{23}$ mol^{-1}.)

1. Calculate the energy of one einstein of light of wavelength 500 nm, in kJ mol^{-1}.
2. A solution transmits 10% of the incident light at some wavelength. Find its absorbance, and the absorbance if the path length is doubled.
3. A photochemical reaction absorbs 2.0×10^{18} photons and reacts 1.0×10^{21} molecules. Find φ and state the mechanism the value suggests.
4. State the two laws of photochemistry and the exact scope of the second (what it does not cover).
5. Distinguish fluorescence from phosphorescence by (a) the states involved, (b) the timescale.
6. A red dye fades rapidly in sunlight while a blue one is fast. Using §1, explain why the sun's UV content, not its brightness in the visible, is usually the culprit — and name the 1–8 scale used to grade this.
7. Why does a UV-absorber finish protect a fibre, in Jablonski terms?
8. A dye solution in the dyebath absorbs strongly but the fibre beneath is being tendered by singlet-oxygen attack sensitised by the dye. Name the process by which the dye's absorbed energy reaches the oxygen.

Answer key

1. $E = N_A hc/\lambda = (6.022 \times 10^{23} \times 6.626 \times 10^{-34} \times 3.00 \times 10^{8}) / (500 \times 10^{-9}) \approx 2.39 \times 10^{5}$ J $mol^{-1} \approx 239$ kJ mol^{-1}.
2. $A = \log (100/10) = 1.0$. Doubling the path doubles A (Lambert–Beer) to 2.0 (transmission 1%).
3. $\varphi = 1.0 \times 10^{21} / 2.0 \times 10^{18} = 500$ — far above 1, indicating a chain reaction.
4. Grotthuss–Draper: only absorbed light can cause photochemical change. Stark–Einstein: one photon activates one molecule in the primary step only — it says nothing about the secondary (dark) reactions that follow, which is how φ can differ from 1.
5. Fluorescence: singlet excited → ground state, nanoseconds. Phosphorescence: triplet excited → ground state (spin-forbidden, hence slow), milliseconds to seconds.
6. Photon energy rises as wavelength falls ($E = hc/\lambda$); near-UV photons carry ~300 kJ mol^{-1} — comparable to bond energies — while visible photons are largely too weak to break the chromophore directly. The grading is the light-fastness scale, 1–8.
7. The absorber takes the photon itself and disposes of the energy by radiationless decay (heat) instead of letting the fibre absorb it and react — it intercepts the Jablonski pathway at the first step.
8. Photosensitisation — energy transfer from the photoexcited dye to oxygen, producing reactive singlet oxygen.